

Exploratory Data Analysis

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2026-02-06

Setting up my environment

Notes: Setting up my environment by loading the 'tidyverse' and related packages

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr       1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(rmarkdown)
library(magrittr)
```

```
##
## Attaching package: 'magrittr'
##
## The following object is masked from 'package:purrr':
##
##   set_names
##
## The following object is masked from 'package:tidyr':
##
##   extract
```

Importing the first 10 rows of the data from a CSV file and assigning it a new variable

```
read.csv("AB_NYC_2019.csv", nrow=10)
```

```
##      id                                name host_id host_name
## 1  2539      Clean & quiet apt home by the park    2787      John
## 2  2595      Skylit Midtown Castle                2845    Jennifer
## 3  3647      THE VILLAGE OF HARLEM....NEW YORK !    4632    Elisabeth
```

```
## 4 3831 Cozy Entire Floor of Brownstone 4869 LisaRoxanne
## 5 5022 Entire Apt: Spacious Studio/Loft by central park 7192 Laura
## 6 5099 Large Cozy 1 BR Apartment In Midtown East 7322 Chris
## 7 5121 BlissArtsSpace! 7356 Garon
## 8 5178 Large Furnished Room Near B'way 8967 Shunichi
## 9 5203 Cozy Clean Guest Room - Family Apt 7490 MaryEllen
## 10 5238 Cute & Cozy Lower East Side 1 bdrm 7549 Ben
## neighbourhood_group neighbourhood latitude longitude room_type
## 1 Brooklyn Kensington 40.64749 -73.97237 Private room
## 2 Manhattan Midtown 40.75362 -73.98377 Entire home/apt
## 3 Manhattan Harlem 40.80902 -73.94190 Private room
## 4 Brooklyn Clinton Hill 40.68514 -73.95976 Entire home/apt
## 5 Manhattan East Harlem 40.79851 -73.94399 Entire home/apt
## 6 Manhattan Murray Hill 40.74767 -73.97500 Entire home/apt
## 7 Brooklyn Bedford-Stuyvesant 40.68688 -73.95596 Private room
## 8 Manhattan Hell's Kitchen 40.76489 -73.98493 Private room
## 9 Manhattan Upper West Side 40.80178 -73.96723 Private room
## 10 Manhattan Chinatown 40.71344 -73.99037 Entire home/apt
## price minimum_nights number_of_reviews last_review reviews_per_month
## 1 149 1 9 2018-10-19 0.21
## 2 225 1 45 2019-05-21 0.38
## 3 150 3 0 NA
## 4 89 1 270 2019-07-05 4.64
## 5 80 10 9 2018-11-19 0.10
## 6 200 3 74 2019-06-22 0.59
## 7 60 45 49 2017-10-05 0.40
## 8 79 2 430 2019-06-24 3.47
## 9 79 2 118 2017-07-21 0.99
## 10 150 1 160 2019-06-09 1.33
## calculated_host_listings_count availability_365
## 1 6 365
## 2 2 355
## 3 1 365
## 4 1 194
## 5 1 0
## 6 1 129
## 7 1 0
## 8 1 220
## 9 1 0
## 10 4 188
```

```
AirBNB_NY <-read.csv("AB_NYC_2019.csv")
```

Exploring the data

```
head(AirBNB_NY)
```

```
## id name host_id host_name
## 1 2539 Clean & quiet apt home by the park 2787 John
## 2 2595 Skylit Midtown Castle 2845 Jennifer
## 3 3647 THE VILLAGE OF HARLEM...NEW YORK ! 4632 Elisabeth
## 4 3831 Cozy Entire Floor of Brownstone 4869 LisaRoxanne
## 5 5022 Entire Apt: Spacious Studio/Loft by central park 7192 Laura
```

```
## 6 5099          Large Cozy 1 BR Apartment In Midtown East      7322      Chris
##   neighbourhood_group neighbourhood latitude longitude      room_type price
## 1          Brooklyn      Kensington 40.64749 -73.97237      Private room  149
## 2          Manhattan      Midtown 40.75362 -73.98377 Entire home/apt  225
## 3          Manhattan      Harlem 40.80902 -73.94190      Private room  150
## 4          Brooklyn Clinton Hill 40.68514 -73.95976 Entire home/apt   89
## 5          Manhattan East Harlem 40.79851 -73.94399 Entire home/apt   80
## 6          Manhattan Murray Hill 40.74767 -73.97500 Entire home/apt  200
##   minimum_nights number_of_reviews last_review reviews_per_month
## 1              1              9 2018-10-19              0.21
## 2              1             45 2019-05-21              0.38
## 3              3              0              NA
## 4              1            270 2019-07-05              4.64
## 5             10              9 2018-11-19              0.10
## 6              3             74 2019-06-22              0.59
##   calculated_host_listings_count availability_365
## 1                             6              365
## 2                             2              355
## 3                             1              365
## 4                             1              194
## 5                             1               0
## 6                             1             129
```

summary (AirBNB_NY)

```
##           id           name           host_id           host_name
## Min.      : 2539   Length:48895   Min.      : 2438   Length:48895
## 1st Qu.: 9471945   Class :character 1st Qu.: 7822033   Class :character
## Median :19677284   Mode  :character Median : 30793816   Mode  :character
## Mean    :19017143
## 3rd Qu.:29152178
## Max.    :36487245
##
## neighbourhood_group neighbourhood      latitude      longitude
## Length:48895      Length:48895      Min.    :40.50   Min.    : -74.24
## Class :character   Class :character 1st Qu.:40.69   1st Qu.: -73.98
## Mode  :character   Mode  :character Median :40.72   Median : -73.96
##                                     Mean  :40.73   Mean  : -73.95
##                                     3rd Qu.:40.76   3rd Qu.: -73.94
##                                     Max.   :40.91   Max.   : -73.71
##
## room_type      price      minimum_nights      number_of_reviews
## Length:48895   Min.      : 0.0   Min.      : 1.00   Min.      : 0.00
## Class :character 1st Qu.: 69.0   1st Qu.: 1.00   1st Qu.: 1.00
## Mode  :character Median : 106.0   Median : 3.00   Median : 5.00
##                                     Mean  : 152.7   Mean  : 7.03   Mean  : 23.27
##                                     3rd Qu.: 175.0   3rd Qu.: 5.00   3rd Qu.: 24.00
##                                     Max.   :10000.0   Max.   :1250.00   Max.   :629.00
##
## last_review      reviews_per_month calculated_host_listings_count
## Length:48895     Min.      : 0.010   Min.      : 1.000
## Class :character 1st Qu.: 0.190   1st Qu.: 1.000
## Mode  :character Median : 0.720   Median : 1.000
##                                     Mean  : 1.373   Mean  : 7.144
##                                     3rd Qu.: 2.020   3rd Qu.: 2.000
```

```
##           Max.      :58.500    Max.      :327.000
##           NA's      :10052
## availability_365
## Min.      : 0.0
## 1st Qu.: 0.0
## Median : 45.0
## Mean      :112.8
## 3rd Qu.:227.0
## Max.      :365.0
##
```

```
glimpse(AirBNB_NY)
```

```
## Rows: 48,895
## Columns: 16
## $ id          <int> 2539, 2595, 3647, 3831, 5022, 5099, 512~
## $ name        <chr> "Clean & quiet apt home by the park", "~
## $ host_id     <int> 2787, 2845, 4632, 4869, 7192, 7322, 735~
## $ host_name   <chr> "John", "Jennifer", "Elisabeth", "LisaR~
## $ neighbourhood_group <chr> "Brooklyn", "Manhattan", "Manhattan", "~
## $ neighbourhood <chr> "Kensington", "Midtown", "Harlem", "Cli~
## $ latitude    <dbl> 40.64749, 40.75362, 40.80902, 40.68514,~
## $ longitude   <dbl> -73.97237, -73.98377, -73.94190, -73.95~
## $ room_type   <chr> "Private room", "Entire home/apt", "Pri~
## $ price       <int> 149, 225, 150, 89, 80, 200, 60, 79, 79,~
## $ minimum_nights <int> 1, 1, 3, 1, 10, 3, 45, 2, 2, 1, 5, 2, 4~
## $ number_of_reviews <int> 9, 45, 0, 270, 9, 74, 49, 430, 118, 160~
## $ last_review  <chr> "2018-10-19", "2019-05-21", "", "2019-0~
## $ reviews_per_month <dbl> 0.21, 0.38, NA, 4.64, 0.10, 0.59, 0.40,~
## $ calculated_host_listings_count <int> 6, 2, 1, 1, 1, 1, 1, 1, 4, 1, 1, 3, ~
## $ availability_365 <int> 365, 355, 365, 194, 0, 129, 0, 220, 0, ~
```

```
## grouping the data by price and removing missing values (NA)
```

```
AirBNB_NY %>%
  group_by(price) %>%
  drop_na ()
```

```
## # A tibble: 38,843 x 16
## # Groups:   price [581]
##       id name      host_id host_name neighbourhood_group neighbourhood latitude
##   <int> <chr>      <int> <chr>      <chr>              <chr>          <dbl>
## 1  2539 "Clean & ~    2787 John      Brooklyn         Kensington     40.6
## 2  2595 "Skylit M~    2845 Jennifer  Manhattan        Midtown        40.8
## 3  3831 "Cozy Ent~    4869 LisaRoxa~ Brooklyn         Clinton Hill   40.7
## 4  5022 "Entire A~    7192 Laura      Manhattan        East Harlem    40.8
## 5  5099 "Large Co~    7322 Chris       Manhattan        Murray Hill    40.7
## 6  5121 "BlissArt~    7356 Garon      Brooklyn         Bedford-Stuy~  40.7
## 7  5178 "Large Fu~    8967 Shunichi    Manhattan        Hell's Kitch~  40.8
## 8  5203 "Cozy Cle~    7490 MaryEllen  Manhattan        Upper West S~  40.8
## 9  5238 "Cute & C~    7549 Ben        Manhattan        Chinatown      40.7
## 10 5295 "Beautifu~    7702 Lena      Manhattan        Upper West S~  40.8
## # i 38,833 more rows
## # i 9 more variables: longitude <dbl>, room_type <chr>, price <int>,
## #   minimum_nights <int>, number_of_reviews <int>, last_review <chr>,
## #   reviews_per_month <dbl>, calculated_host_listings_count <int>,
```

```
## #   availability_365 <int>
```

```
zero <-AirBNB_NY %>%  
  filter(price<1)
```

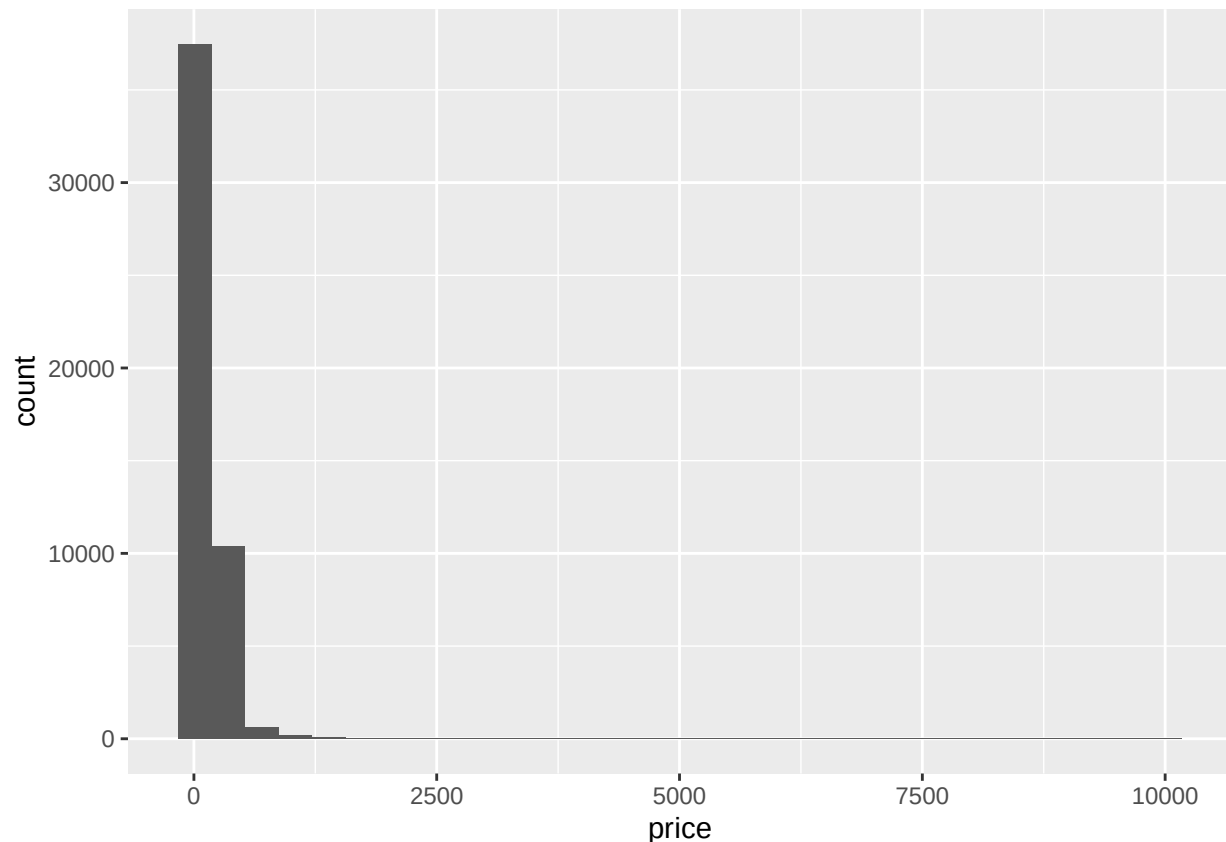
```
filtered_price <-AirBNB_NY %>%  
  filter (price>=1)  
summary (filtered_price$price)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      10.0   69.0   106.0   152.8   175.0  10000.0
```

Plotting a histogram

```
ggplot(data=filtered_price) + geom_histogram(mapping=aes(x=price))
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



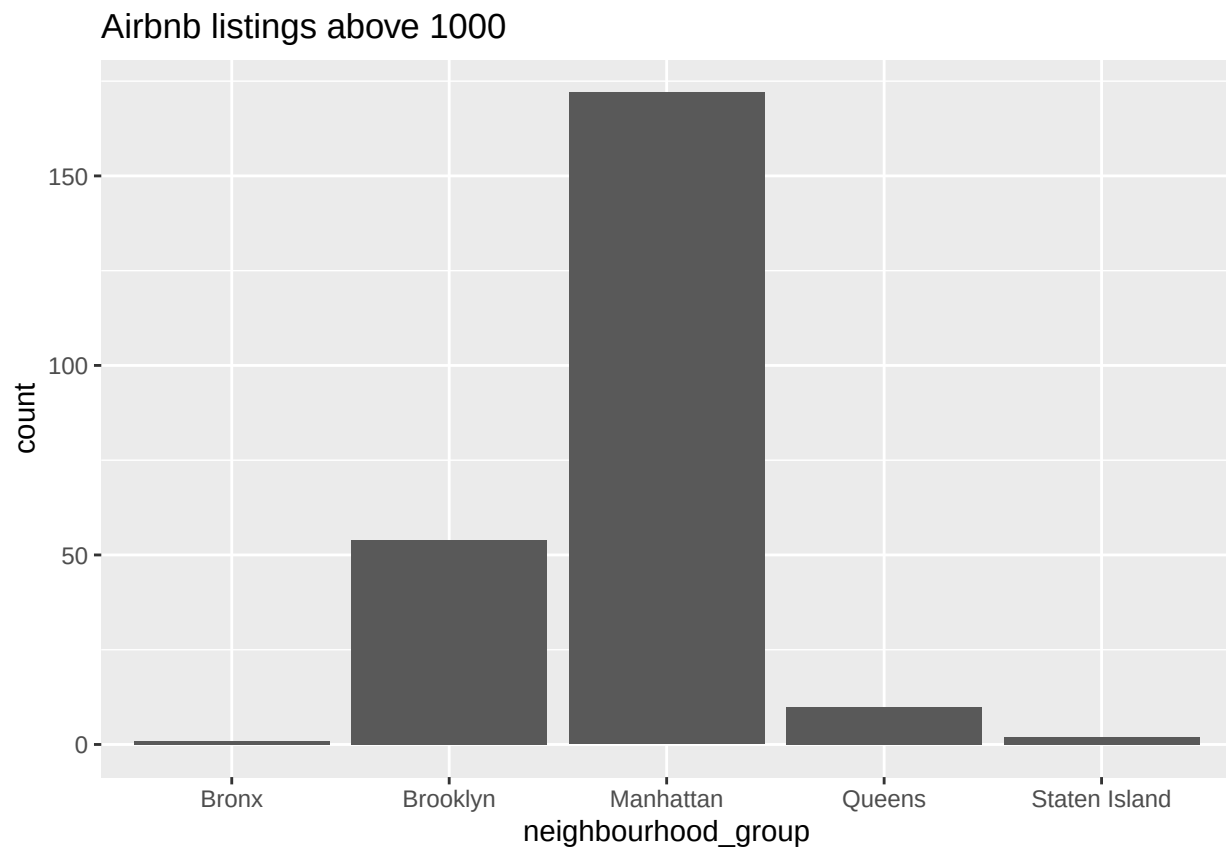
```
# Filtering and assigning a new variable to neighbourhoods with listings above $1000
```

```
above_1000 <-filtered_price %>%  
  filter (price>1000)
```

plotting a bar graph displaying neighbourhood groups with Airbnbs listings above \$1000

```
ggplot(data=above_1000) +  
  geom_bar(mapping=aes(x=neighbourhood_group)) +
```

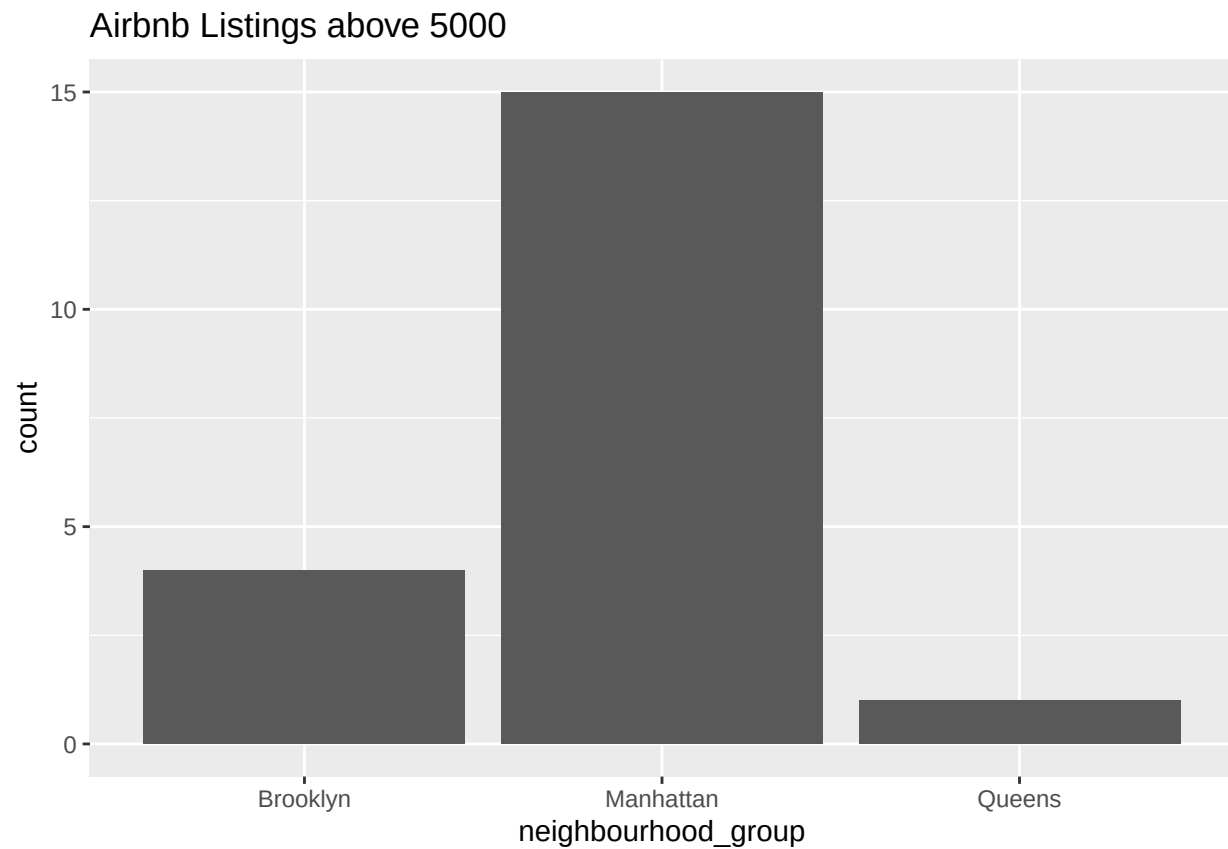
```
labs(title="Airbnb listings above 1000")
```



```
above_5000 <- filtered_price %>%  
  filter (price>5000)
```

plotting a bar graph displaying neighbourhood groups with Airbnbs listings above \$5000

```
ggplot(data=above_5000) +  
  geom_bar(mapping=aes(x=neighbourhood_group)) +  
  labs(title="Airbnb Listings above 5000")
```



```
# Plotting the relationship between price and number of reviews  
ggplot(data=filtered_price)+  
  geom_point(mapping=aes(x=number_of_reviews, y=price, color=neighbourhood_group))+  
  labs(title="Relationship between price and reviews")
```

Relationship between price and reviews

